15 SEP 2022

Roll No.: ______________

Amrita Vishwa Vidyapeetham

Amrita School of Engineering, Bangalore

Computer Science and Engineering

Periodical Test 1, September 2022

Seventh Semester

19CSE353 Mining of Massive Datasets

Duration: 2 hrs **Answer all the questions** **Max. Marks: 50**

CO	Course Outcome
CO1	Understand the importance of data to business decisions, strategy and behavior. Predictive analytics, data mining and machine learning as tools give new methods for analyzing massive data sets
CO2	Expose the students to Big data systems like Hadoop, Spark and Hive.
CO3	Explore means to deal with huge document databases and infinite streams of data to mining large social networks and web graphs. Also, learn Algorithms suitable for large scale mining
CO4	Use of analytic tools, case-studies to provide first hand insight into how big-data problems and their solutions are commercially derived
CO5	Design Large Scale Machine Learning algorithms with Practical hands-on experience for analyzing very large amounts of data

Answer all questions

1.	AirTel consults a data-miner to dig into the call records of the operator. No specific targets are given to the Data Miner. A quantitative target of finding at least 2 new patterns in a month is given. As the data miner starts digging into the data, he finds a pattern that there are fewer international calls on Wednesday than on other days. This information is shared with the management, and they come up with the plan to reduce the international call rates on Wednesdays and start a campaign. Call rates surge, customers are happy with low call prices, more customers sign up, and they make more money! Discuss the various data mining steps used behind the scenes.	[10]	[BTL4] [CO3]
2.A	You plan for a weekend trip to Goa with your friends, you search the internet for right places to visit in Goa. The next time you open the internet, you find ads about various hotels in Goa for staying. How does Datamining play its role in this scenario? Discuss in detail using a relevant DM algorithm.	[6]	[BTL3] [CO2]
2.B	Assume we have a large collection of corpus (50000 documents), count the number of times each 5-word sequence occurs in a large corpus of documents. Write a pseudo code for MapReduce to build a language model	[4]	[BTL2] [CO 2]
3.A	1. Consider a document containing 2000 words, wherein the word hive appears 50 times. Calculate the TF. 2. Assume we have 20 million documents and the word hive appears in 5000 of these. Find the IDF.	[4]	[BTL 2] [CO3]
3.B	Discuss how is a Hash table used to build an Index with a neat diagram.	[6]	[BTL3] [CO 2]

4.A	What are different techniques used for Data Mining? Discuss any two.	[4]	[BTL1] [CO1]
4.B	Distinguish between Hashing and Indexing	[6]	[BTL 2] [CO 2]
5.A	Discuss how do Facebook data centres use MapReduce to process huge volumes of data worldwide. Discuss the technique /steps behind MapReduce	[8]	[BTL 2] [CO3]
5.B	How does Log Transformation help for predicting accuracy in Machine Learning models, Discuss with an example.	[2]	[BTL 2] [CO 2]

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Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**4**	BTL 1	**4**
CO02	**24**	BTL 2	**24**
CO03	**22**	BTL 3	**12**
CO04	**0**	BTL 4	**10**
CO05	**0**	BTL 5	**0**